

2.7 电位的边值问题与解的唯一性

1) 电场强度的唯一性定理与电位的边值问题

一个已知有界区域中的（静）电场的源

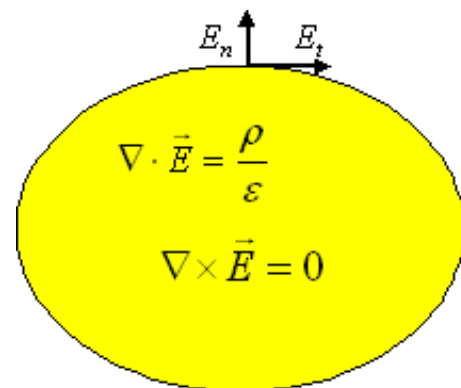
$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon} \quad \nabla \times \vec{E} = 0 \quad \vec{E} = -\nabla \Phi$$

给定电场强度的散度源和旋度源，可以得到：

$$\nabla^2 \Phi = -\frac{\rho}{\varepsilon}$$

电场强度边界条件：

$$E_n = -\frac{\partial \Phi}{\partial n} \quad \int_a^b E_t dl = \Phi(a) - \Phi(b)$$



电场强度边界条件（法向分量或切向分量）对应着电位的边界条件（电位法向方向导数或边界上电位的值）。电位的边值问题（或定解问题）就是求解满足区域**边界条件**的**偏微分方程**（泊松方程或拉普拉斯方程）的定解问题。

2) 电位边值问题的分类

第一类是给定区域边界上的电位值（狄里赫利(Dirichlet)边界条件）

$$\nabla^2 \Phi = -\frac{\rho}{\varepsilon}$$
$$\Phi \Big|_S = f$$

第二类是给定区域边界上的电位的法向导数值（纽曼(Neumann)边界条件）。

$$\nabla^2 \Phi = -\frac{\rho}{\varepsilon}$$
$$\frac{\partial \Phi}{\partial n} \Big|_S = g$$

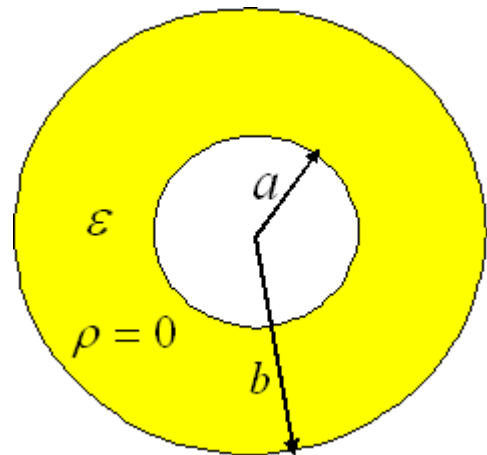
第三类是混合边界条件，在区域的一部分边界上给定电位值，另一部分边界上给定电位的法向导数值。

3) 电位边值问题解唯一的条件

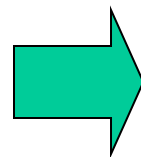
- 静电场电位的**唯一性定理**：当电位满足泊松方程或拉普拉斯方程，在边界上满足三类边界条件之一时，电位的解是唯一的。
- **求解过程**：
 - 解泛定方程的通解；
 - 用边界条件求特解；

例：在球形区域，已知：

- 1) 在 $r=a$ 的球面边界上 $\frac{\partial \Phi}{\partial r} = -\frac{\rho_s}{\epsilon_0}$
- 2) 在 $r=b$ 的球面边界上 $\Phi = 0$
- 3) 在 $a < r < b$ 之间电荷体密度为零；
求 $a < r < b$ 之间的电位。



解： $a < r < b$ $\rho = 0$ $\nabla^2 \Phi = 0$



由题意，电位具有球对称性 $\Phi(r, \theta, \varphi) \rightarrow \Phi(r)$

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Phi(r)}{dr} \right) = 0$$

$$\left. \frac{\partial \Phi}{\partial r} \right|_{r=a} = -\frac{\rho_s}{\epsilon_0}$$

$$\left. \Phi \right|_{r=b} = 0$$

$$\Phi(r) = \frac{c_1}{r} + c_2$$

$$\frac{\partial \Phi}{\partial r} = -\frac{c_1}{r^2} \Rightarrow \left. \frac{\partial \Phi}{\partial r} \right|_{r=a} = -\frac{c_1}{a^2} = -\frac{\rho_s}{\epsilon_0} \Rightarrow c_1 = \frac{a^2 \rho_s}{\epsilon_0}$$

$$\left. \Phi \right|_{r=b} = -\frac{c_1}{b^2} + c_2 = 0 \Rightarrow c_2 = \frac{a^2 \rho_s}{\epsilon_0 b}$$

$$\Phi(r) = \frac{a^2 \rho_s}{\epsilon_0} \left(\frac{1}{r} - \frac{1}{b} \right)$$

2.8 分离变量法

例1：两个无限大接地导体平面，间距 d ，一端接电位为 V 的导体面，求中间区域电位。

解：该问题的边值问题为：

$$\nabla^2 \phi(x, y) = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

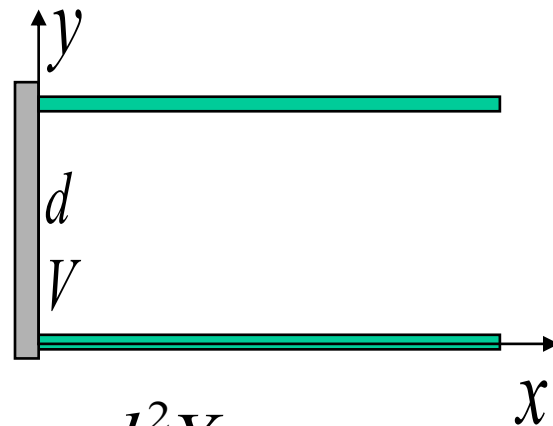
$$\phi(0, y) = V$$

$$\phi(\infty, y) = 0$$

$$\phi(x, 0) = 0$$

$$\phi(x, d) = 0$$

$$\phi(x, y) = X(x)Y(y)$$



$$Y(y) \frac{d^2 X}{dx^2} + X(x) \frac{d^2 Y}{dy^2} = 0$$

$$\Rightarrow \frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} = 0$$

$$\frac{1}{X} \frac{d^2 X}{dx^2} = -k_x^2 \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = -k_y^2$$

$$k_x^2 + k_y^2 = 0$$

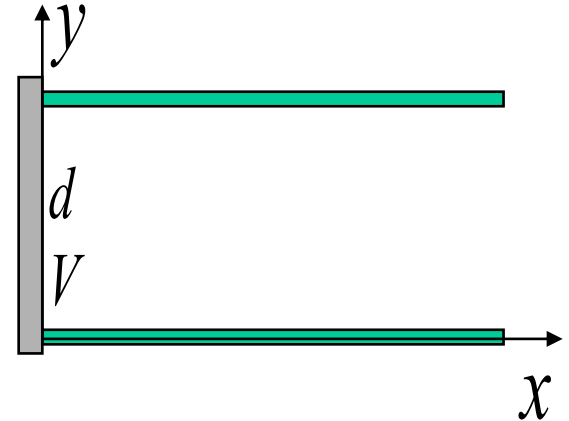
$$\frac{1}{X} \frac{d^2 X}{dx^2} = -k_x^2$$

$$k_x^2 + k_y^2 = 0$$

$$\frac{1}{Y} \frac{d^2 Y}{dy^2} = -k_y^2$$

$$X(x) = Ae^{-jk_x x} + Be^{jk_x x}$$

$$Y(y) = Ce^{-jk_y y} + De^{jk_y y}$$



$$k_x = k \quad X(x) = A \cos kx + B \sin kx$$

$$k_y = jk \quad Y(y) = Ce^{ky} + De^{-ky}$$

$$k_y = k \quad Y(y) = C \sin ky + D \cos ky$$

$$k_x = jk \quad X(x) = Ae^{kx} + Be^{-kx}$$

沿X方向衰减，Y方向周期变化。

$$\phi(x, y) = (Ae^{kx} + Be^{-kx})(C \sin ky + D \cos ky)$$

边界条件:

$$\phi(x, 0) = 0$$

$$\phi(x, 0) = (Ae^{kx} + Be^{-kx})D = 0 \Rightarrow D = 0$$

$$\phi(x, y) = C \sin ky (Ae^{kx} + Be^{-kx})$$

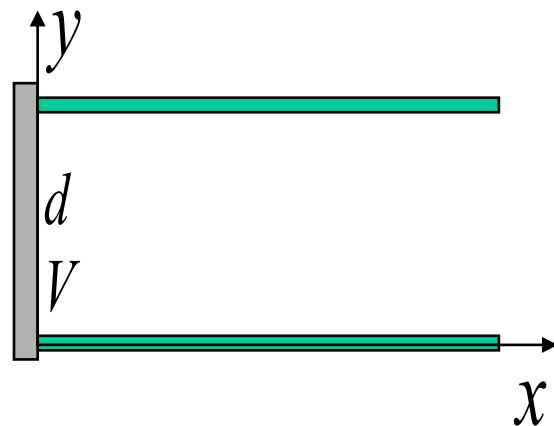
$$\phi(\infty, y) = 0 \Rightarrow A = 0$$

$$\phi(x, y) = C \sin ky \cdot Be^{-kx}$$

$$\phi(x, d) = 0$$

$$\phi(x, d) = (Be^{-kx})C \sin kd = C'e^{-kx} \sin kd = 0 \Rightarrow k = \frac{m\pi}{d} \quad (m = 1, 2, \dots)$$

$$\phi(x, y) = C'e^{-\frac{m\pi}{d}x} \sin \frac{m\pi}{d}y \quad (m = 1, 2, \dots)$$



通解为所有线性无关特解的线性组合：

$$\phi(x, y) = \sum_{m=1}^{\infty} C'_m e^{-\frac{m\pi}{d}x} \sin \frac{m\pi}{d} y$$

$$\phi(0, y) = V$$

$$\phi(0, y) = \sum_{m=1}^{\infty} C'_m \sin\left(\frac{m\pi}{d} y\right) = V$$

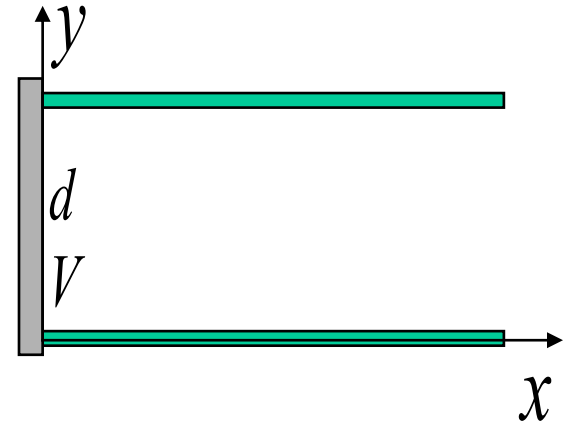
常数V在[0, d]区域的正弦级数展开：

$$V = \sum_{m=1}^{\infty} C'_m \sin \frac{m\pi}{d} y$$

其展开系数为：

$$\begin{aligned} C'_m &= \frac{2}{d} \int_0^d V \sin\left(\frac{m\pi}{d} \xi\right) d\xi = -\frac{2V}{m\pi} \cos\left(\frac{m\pi}{d} \xi\right) \Big|_0^d \\ &= -\frac{2V}{m\pi} [\cos(m\pi) - 1] = \begin{cases} \frac{4V}{m\pi} & m=2n-1 \\ 0 & m=2n \end{cases} \end{aligned}$$

$$\phi(x, y) = \frac{4V}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} e^{-\frac{2n-1}{d}\pi x} \sin \frac{2n-1}{d} \pi y$$



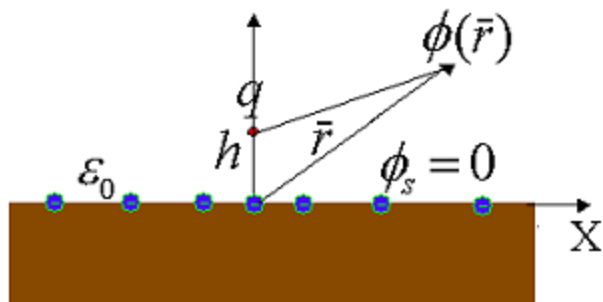
分离变量法解题步骤:

- 根据边界的几何形状和场的分布特征选定坐标系，写出对应的边值问题（微分方程和边界条件）；
- 分离变量，将一个偏微分方程，分离成几个常微分方程；
- 解常微分方程，得到通解；
- 利用给定的边界条件确定积分常数，最终得到电位函数的解。

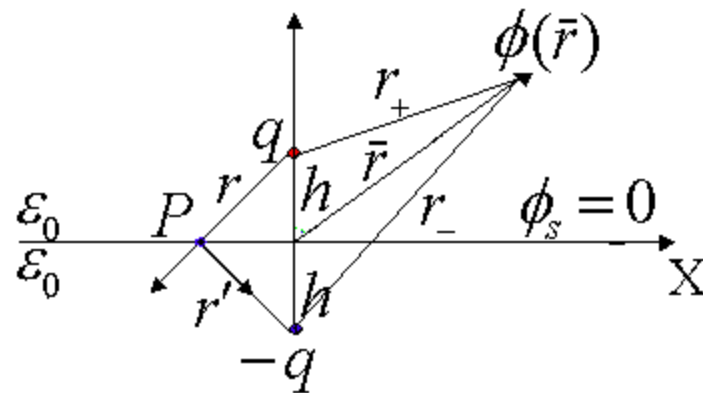
2.9 镜像法

唯一性定理两问题的**等效条件**：研究域内源的分布不变；边界上电位的边界条件不变。

1) 理想导电平面上方的电荷的电场



等效问题为：



等效问题边界上P点的电位为：

$$\phi_s(P) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{r'} \right) = 0$$

与原问题边界条件相同，可等效。

$$r_+ = \sqrt{x^2 + y^2 + (z - h)^2}$$

上半空间任一点R的电位为：

$$\phi(R) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_+} - \frac{1}{r_-} \right)$$

$$r_- = \sqrt{x^2 + y^2 + (z + h)^2}$$

例1. 用无限大的导电平面折成一直角区域，直角区有一点电荷 q 。求直角区域中的电位。

解： 边界条件：

$$\Phi(x=0, y, z) = 0$$

$$\Phi(x, y=0, z) = 0$$

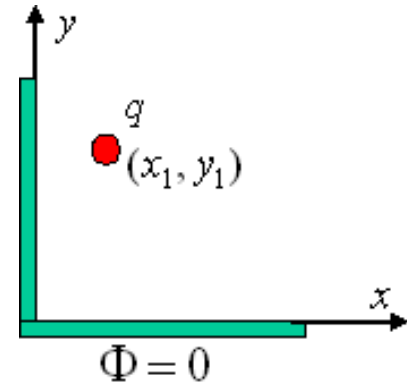
$$\Phi(x, y, z) = \frac{q}{4\pi\epsilon_0 r_1} - \frac{q}{4\pi\epsilon_0 r_2} + \frac{q}{4\pi\epsilon_0 r_3} - \frac{q}{4\pi\epsilon_0 r_4}$$

$$r_1 = \sqrt{(x-x_1)^2 + (y-y_1)^2 + z^2}$$

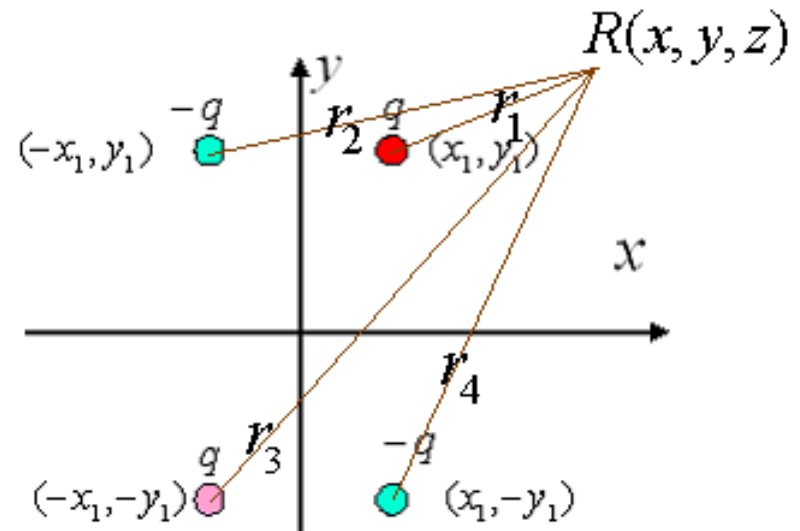
$$r_2 = \sqrt{(x+x_1)^2 + (y-y_1)^2 + z^2}$$

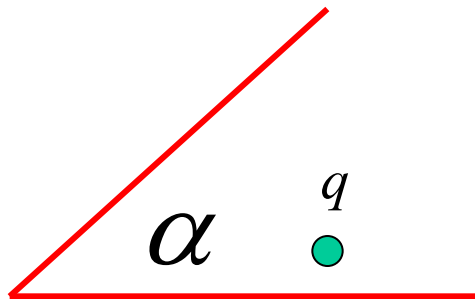
$$r_3 = \sqrt{(x+x_1)^2 + (y+y_1)^2 + z^2}$$

$$r_4 = \sqrt{(x-x_1)^2 + (y+y_1)^2 + z^2}$$



等效问题



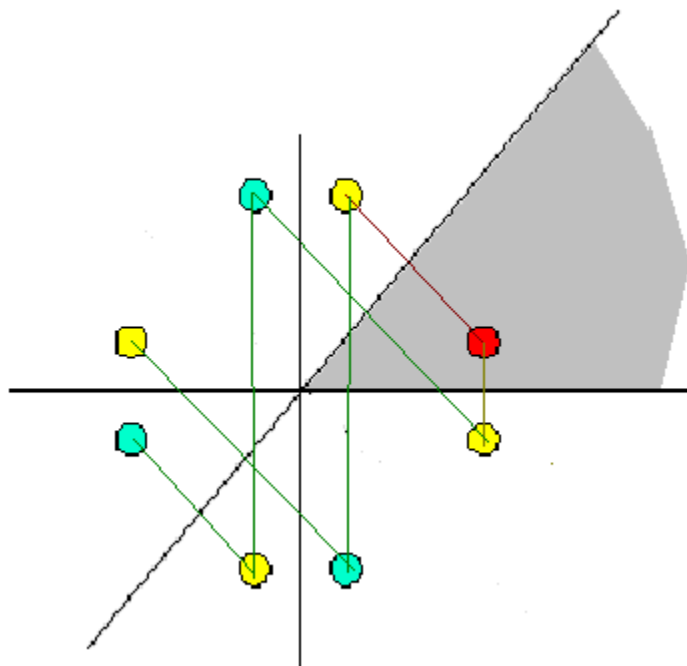


$$\alpha = \frac{\pi}{n}$$



镜像电荷数: $2n-1$

例: $\alpha = 45^\circ$



2) 导体球附近点电荷的电场

边界条件:

$$\Phi(r = a) = 0$$

要使边值问题相同
球面上的电位应为零

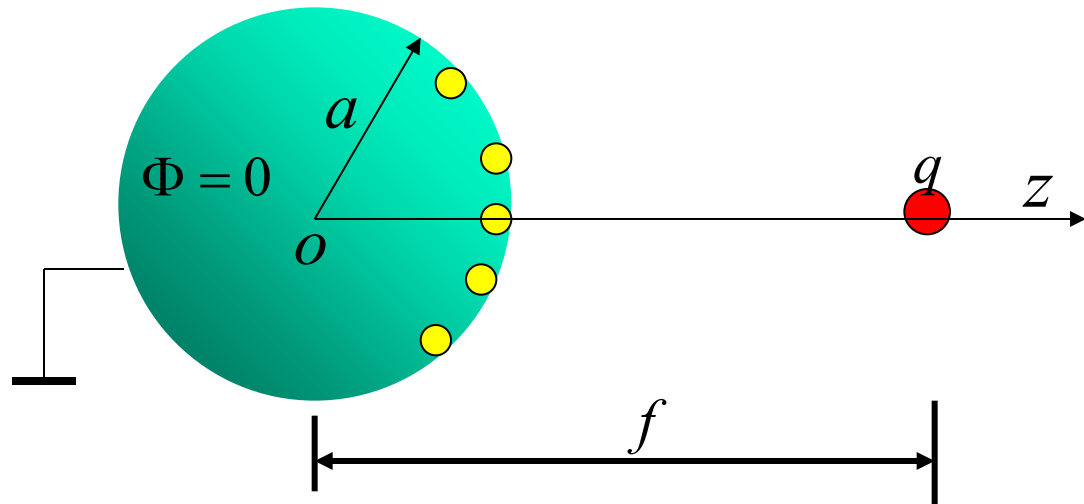
$$\Phi(r = a) = \frac{q}{4\pi\epsilon_0 r_1} + \frac{q'}{4\pi\epsilon_0 r_2} = 0$$

$$\frac{q}{r_1} + \frac{q'}{r_2} = 0 \quad \Rightarrow \quad \frac{r_2}{r_1} = \frac{-q'}{q}$$

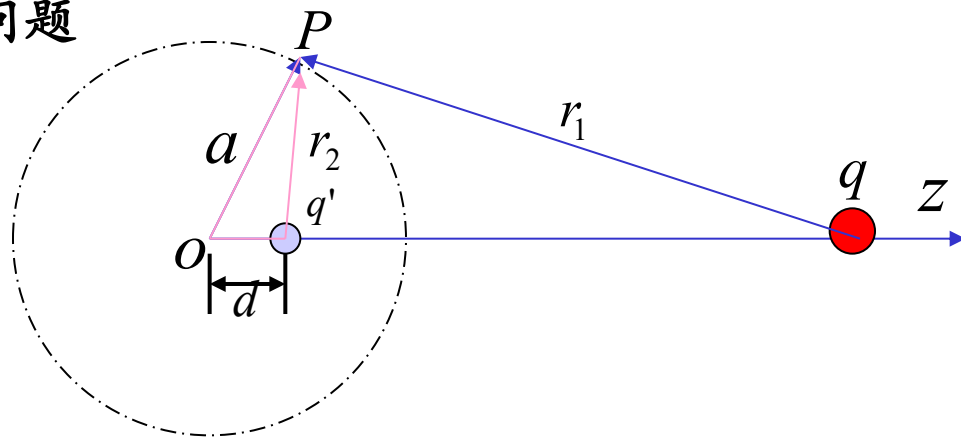
$$\Delta o P q \propto \Delta o q' P \quad \Rightarrow \quad \frac{r_2}{r_1} = \frac{a}{f} = \frac{d}{a}$$

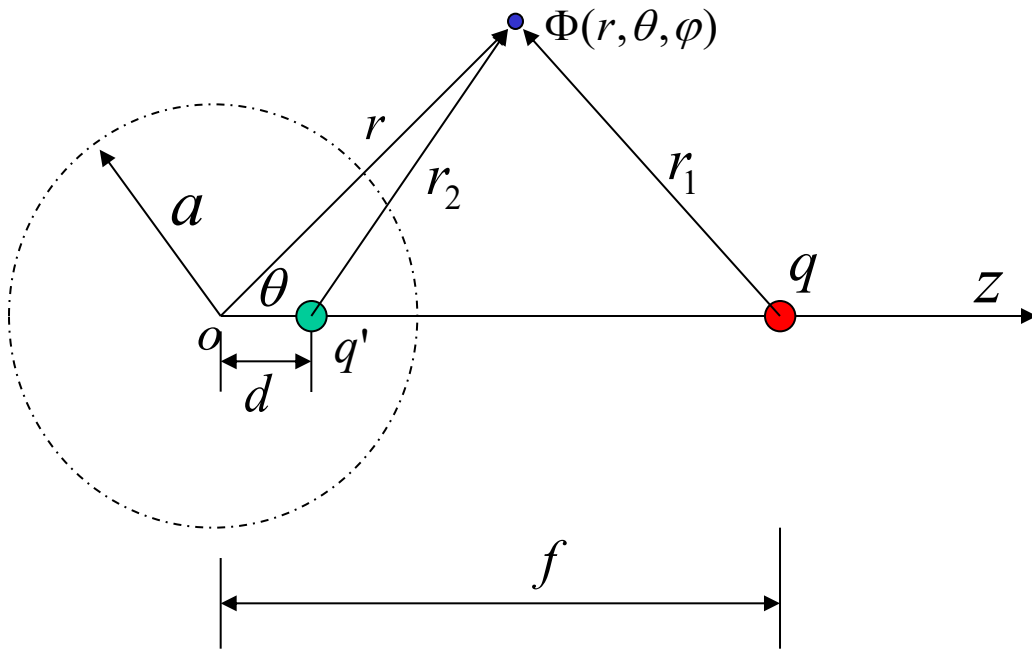
$$d = \frac{a^2}{f}$$

$$q' = -\frac{a}{f} q$$



等效问题





$$d = \frac{a^2}{f}$$

$$q' = -\frac{a}{f}q$$

导体球附近点电荷的电位为

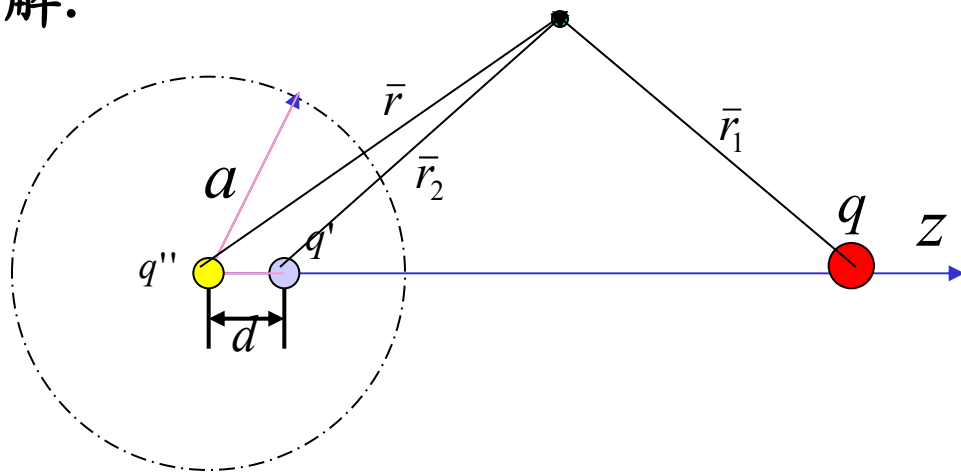
$$\Phi(r, \theta, \varphi) = \frac{q}{4\pi\epsilon_0 r_1} + \frac{q'}{4\pi\epsilon_0 r_2}$$

$$r_1 = \sqrt{r^2 + f^2 - 2rf \cos \theta}$$

$$r_2 = \sqrt{r^2 + d^2 - 2rd \cos \theta}$$

例2 导体球半径为 a , 电位为 V , 距离导体球中心为 f 处放置电量为 q 的点电荷。
求电位。

解:

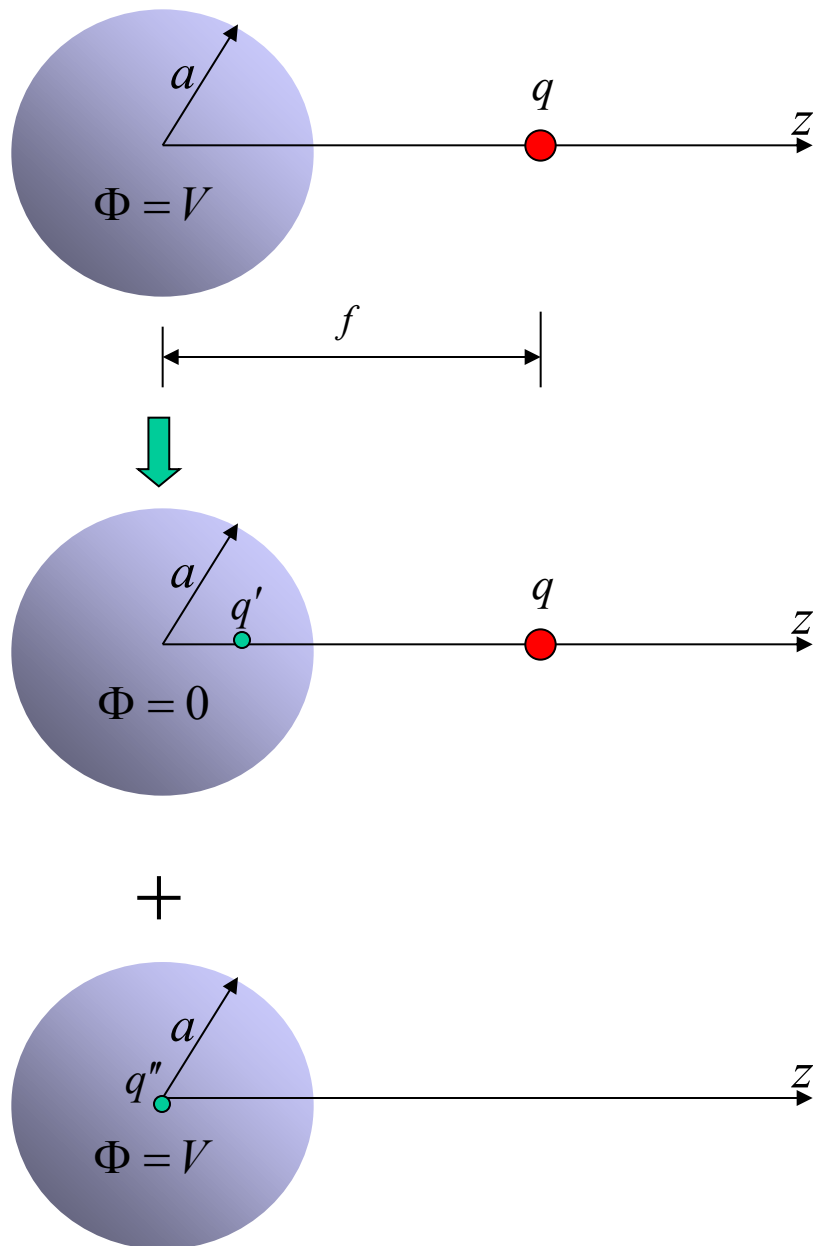


$$\Phi = \Phi_1 + \Phi_2 + \Phi_3$$

$$\Phi_a = \Phi_1 + \Phi_2 = \frac{q}{4\pi\epsilon_0 r_1} + \frac{q'}{4\pi\epsilon_0 r_2}$$

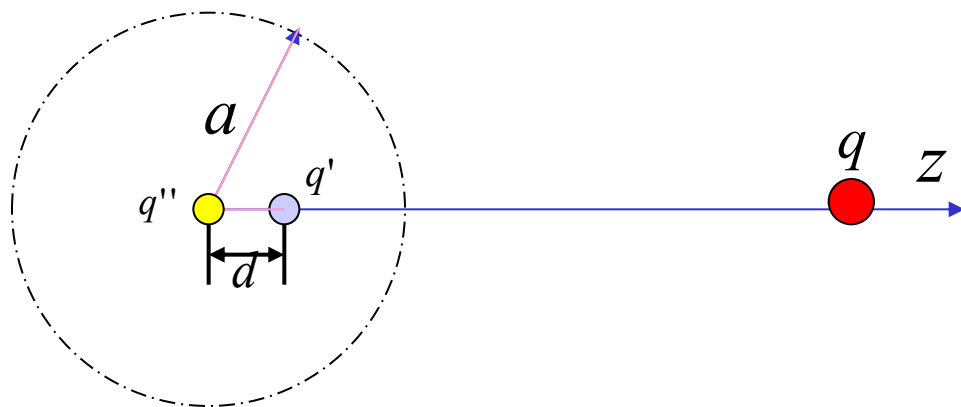
$$V = \frac{q''}{4\pi\epsilon_0 a} \Rightarrow \frac{q''}{4\pi\epsilon_0} = Va$$

$$\Phi_b = \Phi_3 = \frac{q''}{4\pi\epsilon_0 r} = \frac{Va}{r}$$



例3 导体球半径为 a ,带电量为 Q ,距离导体球中心为 f 处放置电量为 q 的点电荷. 求电位.

解:

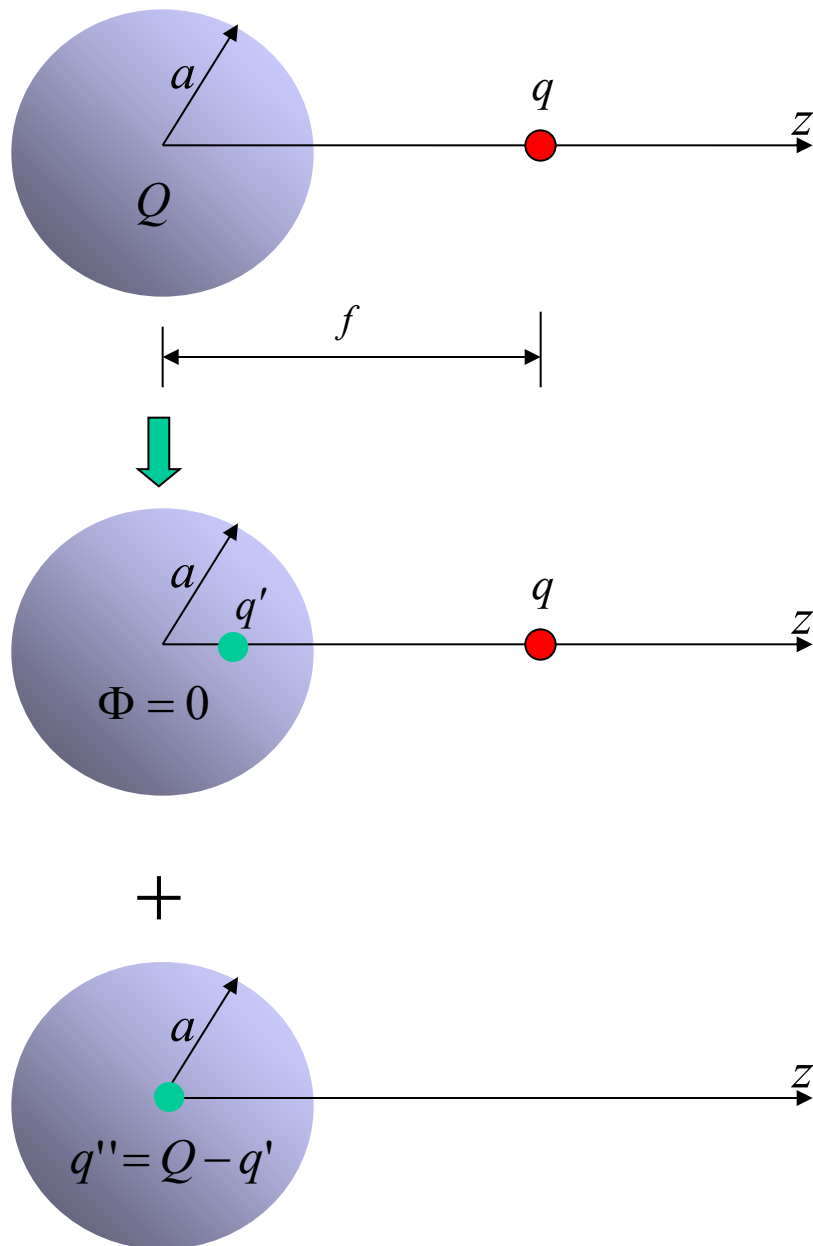


$$\Phi = \Phi_1 + \Phi_2 + \Phi_3$$

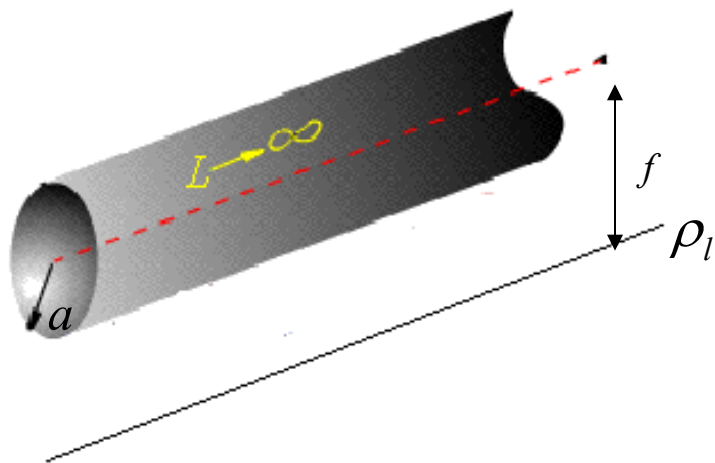
$$\Phi_a = \Phi_1 + \Phi_2 = \frac{q}{4\pi\epsilon_0 r_1} + \frac{q'}{4\pi\epsilon_0 r_2}$$

$$\Phi_b = \Phi_3 = \frac{q''}{4\pi\epsilon_0 r}$$

$$Q = q' + q''$$

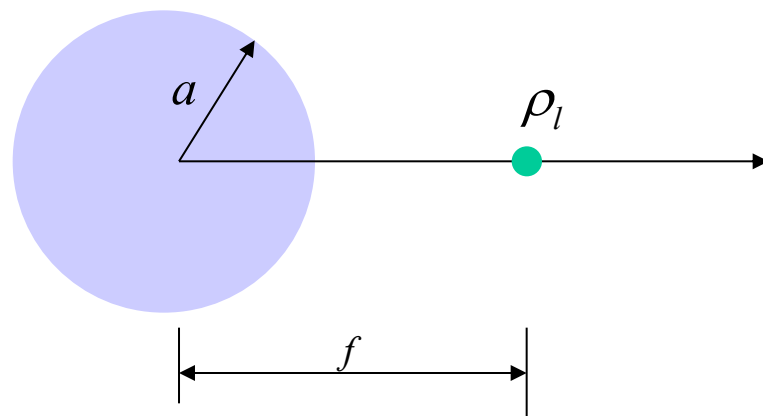


3) 无限长导体圆柱附近有平行放置的线电荷的电位



要使边值问题相同
圆柱面上的电位应为常数

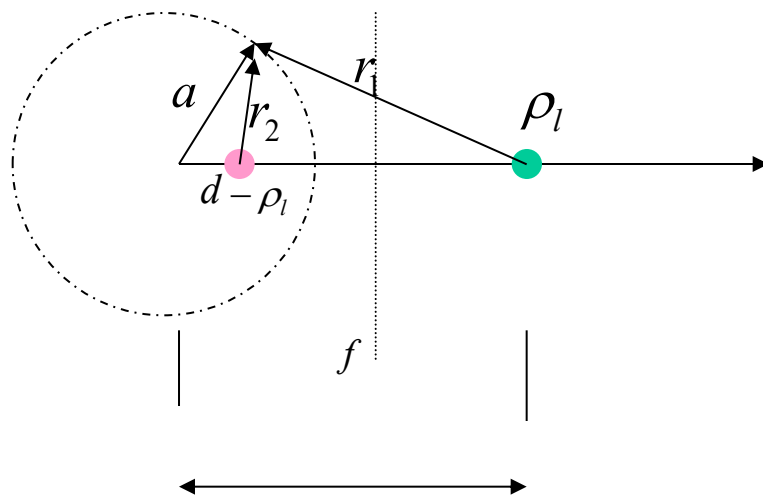
等效问题



$$\vec{E}(\vec{r}) = \frac{\rho_l}{2\pi\epsilon_0\rho} \hat{\rho} \quad \Phi(\vec{r}) = \int_r^{r_0} \frac{\rho_l}{2\pi\epsilon_0 r} dr = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{r_0}{r}$$

$$\Phi_s = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{r_0}{r_1} - \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{r_0'}{r_2} = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{r_0 r_2}{r_1 r_0'}$$

为计算方便，选电位参考点在线电荷
与其镜像电荷之间的中点 $r_0 = r_0'$



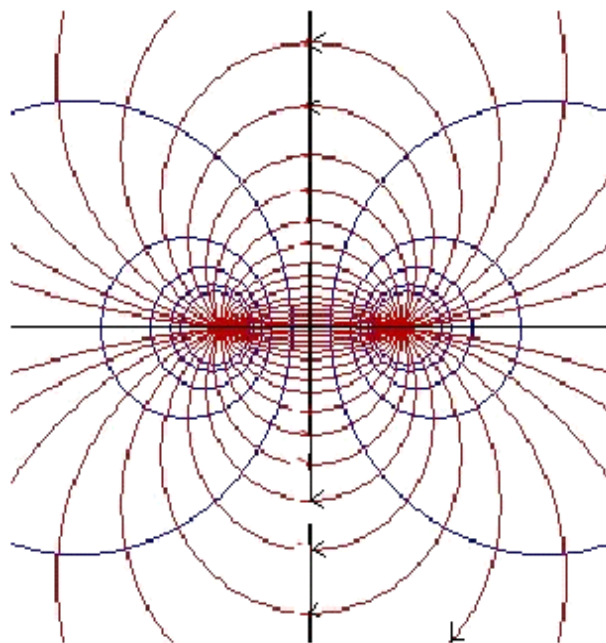
$$\Phi_s = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{r_2}{r_1} = C \quad \frac{r_2}{r_1} = C \quad \frac{r_2}{r_1} = \frac{a}{f} = \frac{d}{a} \quad d = \frac{a^2}{f} \quad \Phi_s = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{a}{f}$$

无限长导体圆柱附近有平行放置的线电荷的电位为

$$\Phi(\vec{r}) = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{r_2}{r_1}$$

线电荷与镜像线电荷具有对称性

$$\Phi_s = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{a}{f} \quad \Phi'_s = \frac{-\rho_l}{2\pi\epsilon_0} \ln \frac{a}{f} = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{f}{a}$$



例4. 一对平行导线，间距为 D ，导线半径为 a 。如果导线间电压为 V ，求电位分布。

解：用镜像法，在其柱面内各放置等量异号的线电荷

线电荷与两根导线的中心距离满足下式

$$f + d = D \quad d = \frac{a^2}{f}$$

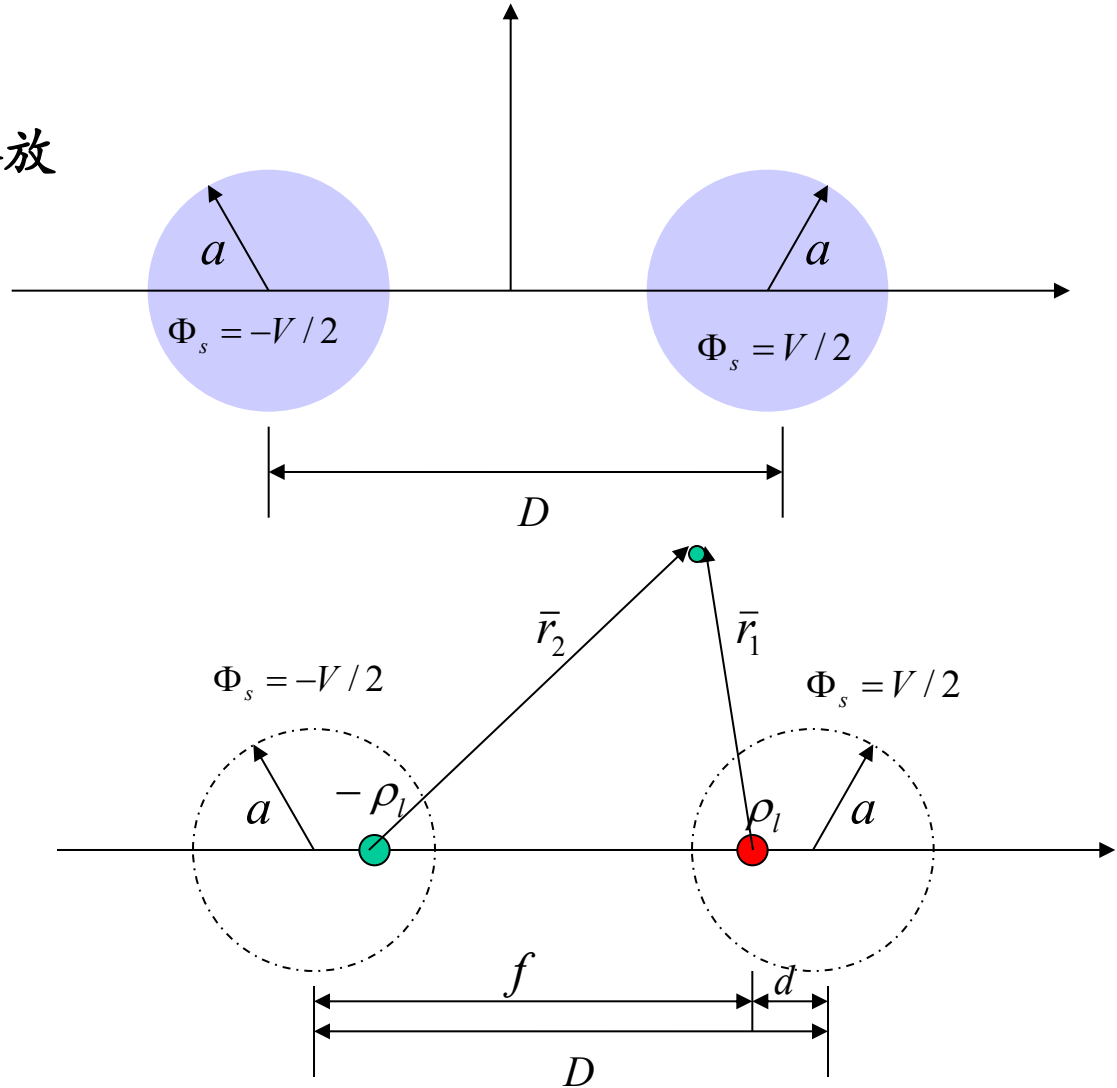
$$f = \frac{D + \sqrt{D^2 - 4a^2}}{2}$$

$$d = \frac{D - \sqrt{D^2 - 4a^2}}{2}$$

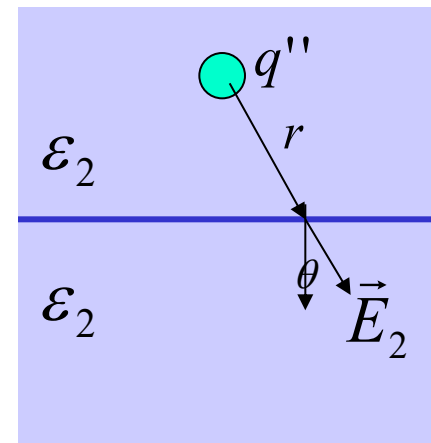
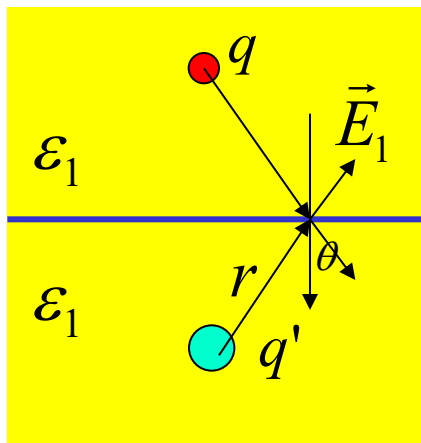
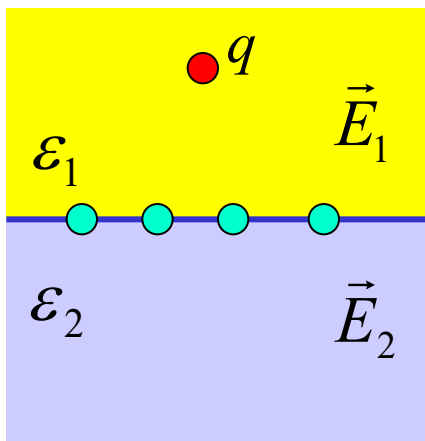
$$\Phi_s = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{f}{a} = \frac{V}{2}$$

$$\rho_l = \frac{\pi\epsilon_0 V}{\ln \frac{D + \sqrt{D^2 - 4a^2}}{2a}}$$

$$\Phi(r, \varphi) = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{r_2}{r_1}$$



4) 无限大介质平面上点电荷的电场



如何计算 q' 和 q'' ?

q' 和 q'' 产生的电场满足边界条件

$$\begin{aligned}
 D_{1n} &= D_{2n} \\
 \Phi_1 &= \Phi_2
 \end{aligned}
 \left\{ \begin{aligned}
 D_{1n} &= \frac{q}{4\pi r^2} \cos \theta - \frac{q'}{4\pi r^2} \cos \theta \\
 D_{2n} &= \frac{q''}{4\pi r^2} \cos \theta \\
 \Phi_1 &= \frac{q}{4\pi \epsilon_1 r} + \frac{q'}{4\pi \epsilon_1 r} \\
 \Phi_2 &= \frac{q''}{4\pi \epsilon_2 r}
 \end{aligned} \right.
 \begin{cases}
 q - q' = q'' \\
 \frac{q + q'}{\epsilon_1} = \frac{q''}{\epsilon_2}
 \end{cases}$$

$$q' = \frac{\epsilon_1 - \epsilon_2}{\epsilon_1 + \epsilon_2} q$$

$$q'' = \frac{2\epsilon_2}{\epsilon_1 + \epsilon_2} q$$

例5. 在 $z>0$ 的上半空间为空气，在 $z<0$ 的下半空间为介质，在空气中 $z=h$ 处有一个点电荷 q 。求此点电荷所受的力。

解： 点电荷受到界面上束缚电荷的作用力，而界面上束缚电荷在上半空间产生的场可通过镜像电荷计算

$$q' = \frac{\epsilon_0 - \epsilon}{\epsilon_0 + \epsilon} q$$

点电荷所受的力

$$\begin{aligned}\vec{F} &= q\vec{E} = q \frac{q'}{4\pi\epsilon_0(2h)^2} \hat{z} \\ &= -\hat{z} \frac{q^2}{16\pi\epsilon_0 h^2} \frac{\epsilon - \epsilon_0}{\epsilon + \epsilon_0}\end{aligned}$$

